Utrecht University

Detective's Evidence Board for Knowledge Graph Visual Analytics

Design Challenge: Visual Analytics for Big Data

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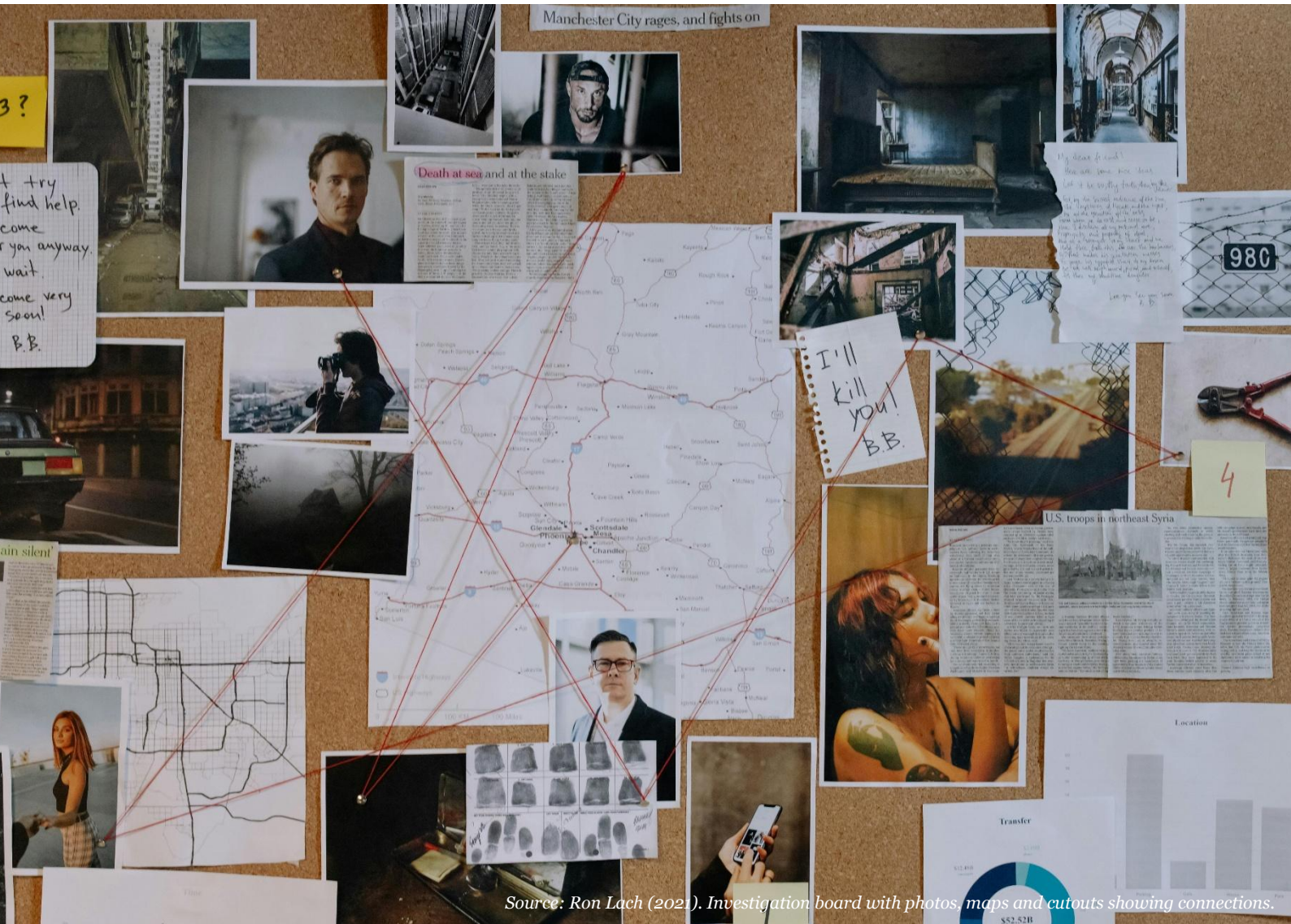

Source: Ron Lach (2021). Investigation board with photos, maps and cutouts showing connections.

The Investigation Begins

A knowledge graph rarely tells you the whole story. It tells you fragments of one, assembled from imperfect sources where relationships may be inferred rather than confirmed, and where the absence of a connection can be just as revealing as its presence. This is the fundamental tension at the heart of the Design Challenge: *how do you build a visual analytics interface that helps a domain expert make sense of a graph that is simultaneously too large to see all at once and too incomplete to trust entirely?*

My answer is EvidenceBoard, a three-view interactive system built on the metaphor of the detective's evidence board, formalised as a computational tool and implemented in D3.js v7 within a Marimo reactive Python environment. This post documents the design process from first principles to working implementation, grounding every decision in established visualisation research and reflecting on what the system achieves and where it falls short.

The Dataset and the Problem

The MC3 knowledge graph describes a fictional but richly structured coastal investigation set in Oceanus, October 2040. The graph contains 1,159 nodes and 3,226 edges. Its structure is tripartite: Entity nodes (people, vessels, organisations, groups, locations), Relationship nodes encoding semantic connection types, and Communication nodes embedding timestamped message events. Five entity subtypes appear across 44 entities retained after filtering, and eight relationship subtypes: Suspicious, Colleagues, Friends, Operates, AccessPermission, Coordinates, Reports, and Unfriendly. Communication nodes carry ISO timestamps spanning October 1 to 14, 2040, enabling 14-day temporal analysis at daily and hourly granularity.

Three analytical tasks frame the challenge. Discovery means identifying new relationships or patterns not immediately obvious from the raw graph. Anomaly detection means finding entities or relationships that violate expected norms, such as an entity that is simultaneously publicly trusted and privately suspected. Missing data inference means predicting where the graph is incomplete based on indirect evidence such as communication frequency between structurally unlinked entities. These tasks are not independent. They co-occur within a single investigative workflow, and the design had to support all three without requiring the analyst to switch modes explicitly.

Two computed properties are novel to this design. Conflict entity detection identifies entities simultaneously holding Suspicious and Colleagues or Friends relationships, computed as a set intersection across relationship subtypes per entity. Ghost link detection identifies entity pairs with five or more direct communications but no corresponding Relationship node in the graph, representing predicted data gaps. Three ghost links exist in the filtered MC3 graph.

What the Research Told Me to Do Differently

I structured the literature review around four themes that directly shaped design decisions. Where papers disagree, I surfaced those tensions because they reveal genuine trade-offs rather than a false consensus.

Design methodology: the nested model as a constraint, not a recipe

Munzner's (2009) four-level nested model couples visualisation design to validation. The levels run from domain problem characterisation through task abstraction, visual encoding, and algorithm design, with each level's output cascading as input to the next. Munzner explicitly identifies Level 2, abstraction, as often the hardest to get right. In designing EvidenceBoard, the initial instinct was to build a better node-link diagram. It was only after articulating the investigative task at Level 1 that the need for three coordinated views became clear. Sedlmair, Meyer, and Munzner (2012) add the practical dimension through their 32 documented pitfalls. Pitfall PF-25, warning against treating a usage scenario as a validation case study, directly shaped how the analyst walkthrough in this post is framed: it is a plausible scenario grounded in the data, not an empirical result.

Task abstraction: overview-first challenged, then partially defended

Shneiderman's (1996) overview-first mantra remains foundational for EvidenceBoard's three-tab architecture. However, Li et al. (2024) conducted interview research with 19 knowledge graph practitioners across eight organisations and found that practitioners rejected overview-first approaches entirely, describing node-link diagrams as great slide decoration. Li et al. identify organic discovery, Wikipedia-style entity-hopping from a known starting point, as the dominant mode of knowledge graph investigation in practice. EvidenceBoard's auto-surfaced highest-risk entity on load is a direct response to this finding: the system does not wait for the analyst to orient themselves but immediately proposes an analytically grounded starting node.

Brehmer and Munzner (2013) refine Shneiderman's flat taxonomy into a multi-level typology structured around three dimensions: why (motivation), how (method), and what (input and output). For EvidenceBoard, the primary goal is Discover, achieved through Browse and Explore interactions, using Navigate (zoom and pan), Filter (entity type and evidence threshold), and Select (entity card click) operations. The distinction between means and ends matters here because the same low-level interaction can serve different analytical purposes depending on context.

Perceptual encoding: Ware's hierarchy as a constraint on every channel choice

Ware (2004) grounds visualisation design in the perceptual psychology of the human visual system. His three-stage processing model separates Stage 1 parallel feature extraction, under 200ms and automatic, from Stage 2 pattern perception via Gestalt principles and Stage 3 sequential goal-directed processing constrained by working memory of approximately four visual chunks. His effectiveness hierarchy for quantitative channels, running from position to

length to area to colour saturation, directly informed the decision to encode evidence count using stroke width rather than saturation. Colour hue is an identity channel with a practical limit of approximately five to seven discriminable categories, directly constraining the entity type palette. The connectedness Gestalt principle, which Ware identifies as the strongest grouping cue, justifies the use of lines rather than any other visual connector in the graph view.

Uncertainty representation: MacEachren's empirical findings over received wisdom

MacEachren et al. (2012) rated colour saturation as unacceptable for uncertainty encoding, below 4.0 out of 7.0. This single empirical finding ruled out the most commonly recommended visual variable in the uncertainty literature and forced the design toward opacity and dashed texture instead.

MacEachren, Roth, O'Brien, Li, Swingley, and Gahegan (2012) provide the only empirical study in this review, comprising two linked experiments testing which visual variables are most intuitively understood for representing uncertainty. Their specific findings: fuzziness received a mean intuition rating of 6.7 out of 7.0, and colour value or lightness received 5.3 out of 7.0. Colour saturation, frequently recommended in prior literature, was rated unacceptable at below 4.0. The dashed line texture used for ghost links follows established cartographic convention for uncertain or provisional features, exploiting a widely understood visual convention rather than introducing a novel encoding. The directional asymmetry identified by MacEachren et al. is respected in EvidenceBoard's opacity encoding: lower opacity encodes lower confidence, which is the natural perceptual reading of the channel.

Investigative analytics: Jigsaw's finding that software cannot replace reading

Stasko, Gorg, and Liu (2007) present Jigsaw, grounded in Pirolli and Card's sensemaking model. Its formal evaluation produced a key finding: software functions cannot replace the reading of reports, and careful reading of message content remained the preferred analytical method even when sophisticated automated tools were available. This directly influenced EvidenceBoard's decision to make communication message text immediately accessible within the evidence chain panel rather than summarising it. Bludau et al. (2023) add a caution: externalising edge detail into separate panels may harm the fluidity of visual data exploration. This is a genuine tension that EvidenceBoard does not fully resolve, as the evidence chain panel requires the analyst to leave the main graph canvas. The Jigsaw finding that reading content remains essential makes this trade-off defensible.

These four themes are not independent strands of literature. Together, they form a coherent design constraint system. Munzner (2009) and Sedlmair et al. (2012) define the process for principled design and its validation. Shneiderman (1996) and Li et al. (2024) define the entry point strategy and reveal a genuine tension between overview-first and seeded discovery. Ware (2004) and Brehmer and Munzner (2013) define which channels and interactions are available

and why. MacEachren et al. (2012) define how uncertainty must be encoded, ruling out the most commonly recommended variable. Stasko et al. (2007) and Bludau et al. (2023) define the trade-off between evidence access and exploration fluidity.

EvidenceBoard is the specific design that emerges from holding all five constraints simultaneously. No prior knowledge graph visualisation system reviewed here treats edge absence as a computationally derived, visually encoded first-class signal, and none provides a formalised mechanism for identifying entities whose public and private relationship profiles are in direct conflict. These two gaps, one structural and one semantic, define the specific contribution this design makes to the field.

Design Process

The design process followed the funnel described by Sedlmair et al. (2012): from a broad consideration space of possible visual idioms, through a proposal space of evaluated candidates, to a satisficing selection (choosing the best available option given the constraints, rather than a theoretically optimal one). Every rejection was grounded in specific literature, not in aesthetic preference.

V0: Adjacency Matrix (Rejected) at Task Abstraction

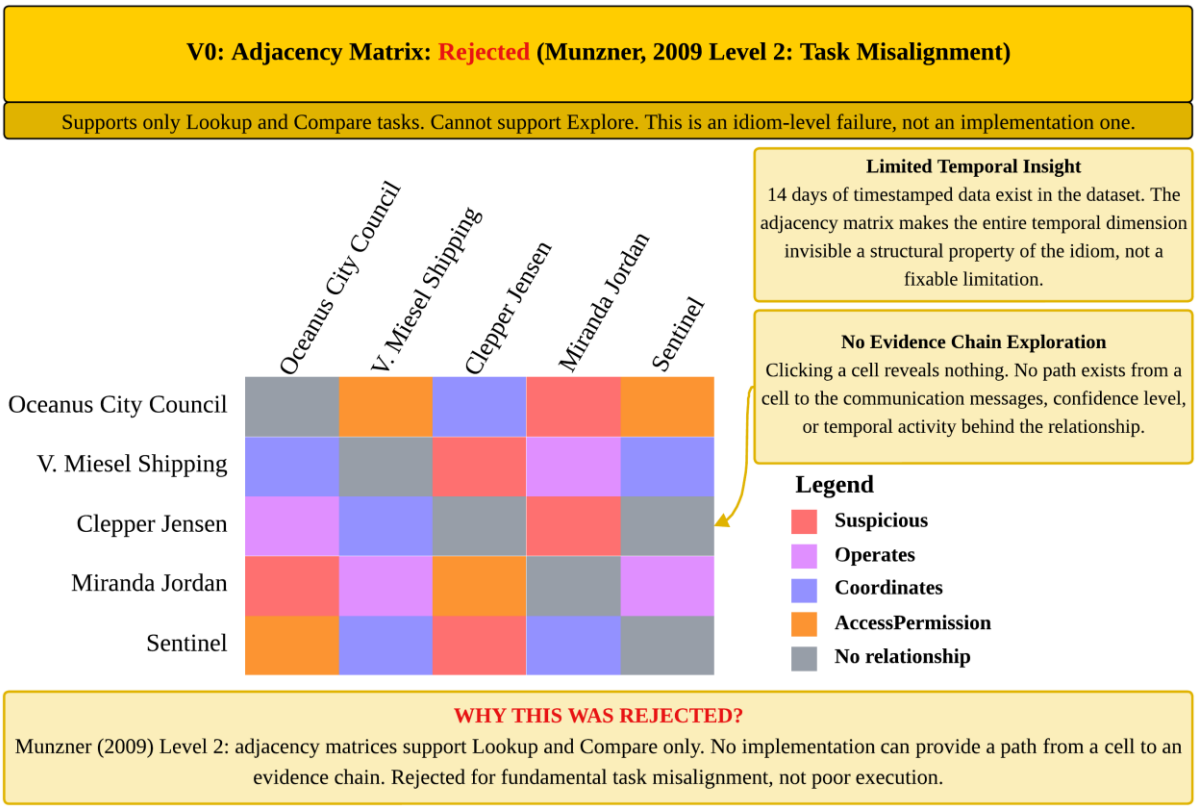

Figure 1a. V0: Adjacency matrix. Rejected at Munzner (2009) Level 2. Supports Lookup and Compare only. No evidence chain access. Temporal dimension invisible.

The first candidate was a standard adjacency matrix encoding entity pairs as cells, with colour value representing relationship strength and colour hue indicating relationship type. Adjacency

matrices eliminate edge crossings entirely and scale well for dense graphs. The rejection reason was task misalignment at Munzner's Level 2. Adjacency matrices support Lookup and Compare tasks but cannot support the Explore task that investigative analysis requires. This is an idiom-level failure: no implementation of an adjacency matrix can provide a path from a cell to the evidence chain behind it. The temporal dimension across 14 investigation days is entirely invisible. The rejection is not about poor execution but about fundamental task misalignment with the analytical goals of the investigation.

V1: Hierarchical Edge Bundling (Rejected) for Violating Uncertainty Encoding

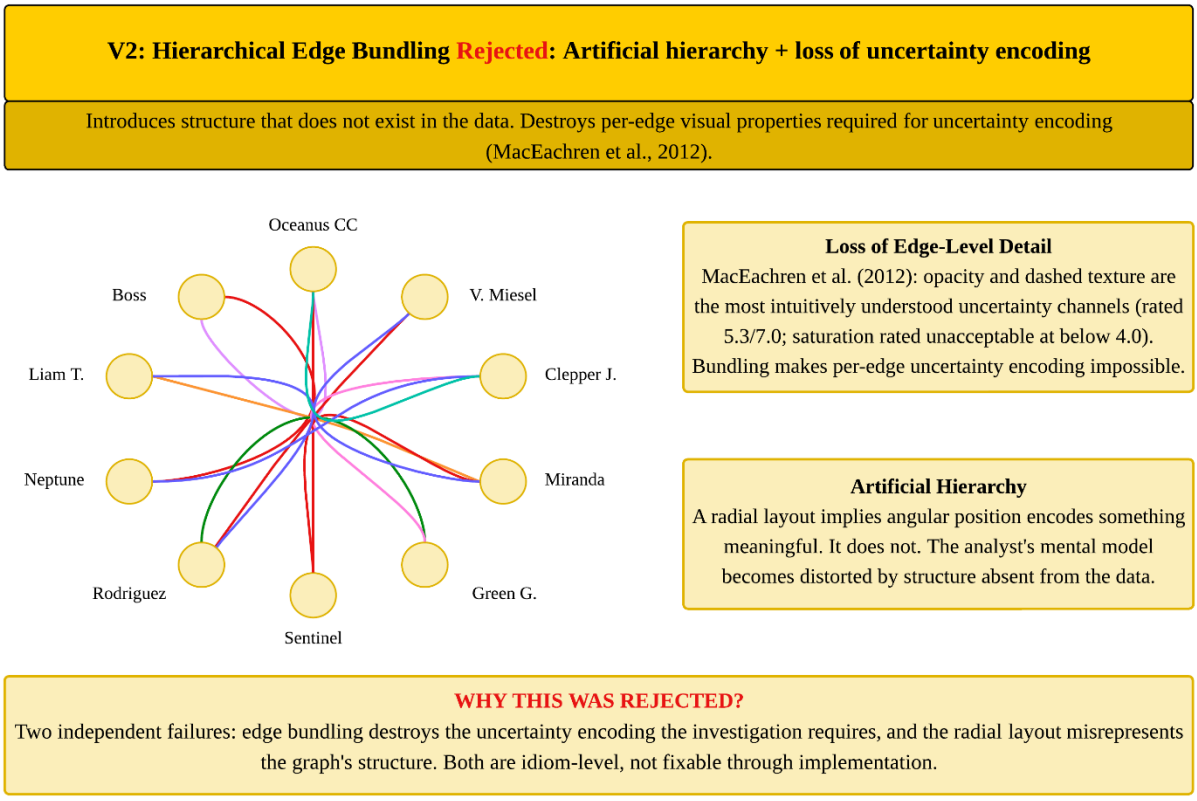

Figure 1b. V1: Hierarchical edge bundling. Rejected for artificial hierarchy and destruction of per-edge uncertainty encoding (MacEachren et al., 2012).

The second candidate used hierarchical edge bundling with entities in a radial layout sorted by subtype. Two rejection reasons emerged. First, the radial layout imposes an artificial hierarchy on a non-hierarchical graph: placing entities on a circle implies that angular position encodes something meaningful, which it does not. The analyst's mental model of the network becomes distorted by a spatially implied structure absent from the data. Second, and more critically, edge bundling destroys the per-edge properties that MacEachren et al. (2012) identified as essential for uncertainty encoding. When edges merge into shared paths, it becomes impossible to assign individual opacity values for confidence or dashed textures for predicted missing connections. The uncertainty encoding that is the most novel contribution of EvidenceBoard cannot exist within this idiom. Both failures are idiom-level, not fixable through implementation.

V2: Bare Force Graph (Rejected) at Sedlmair Pitfall PF-32

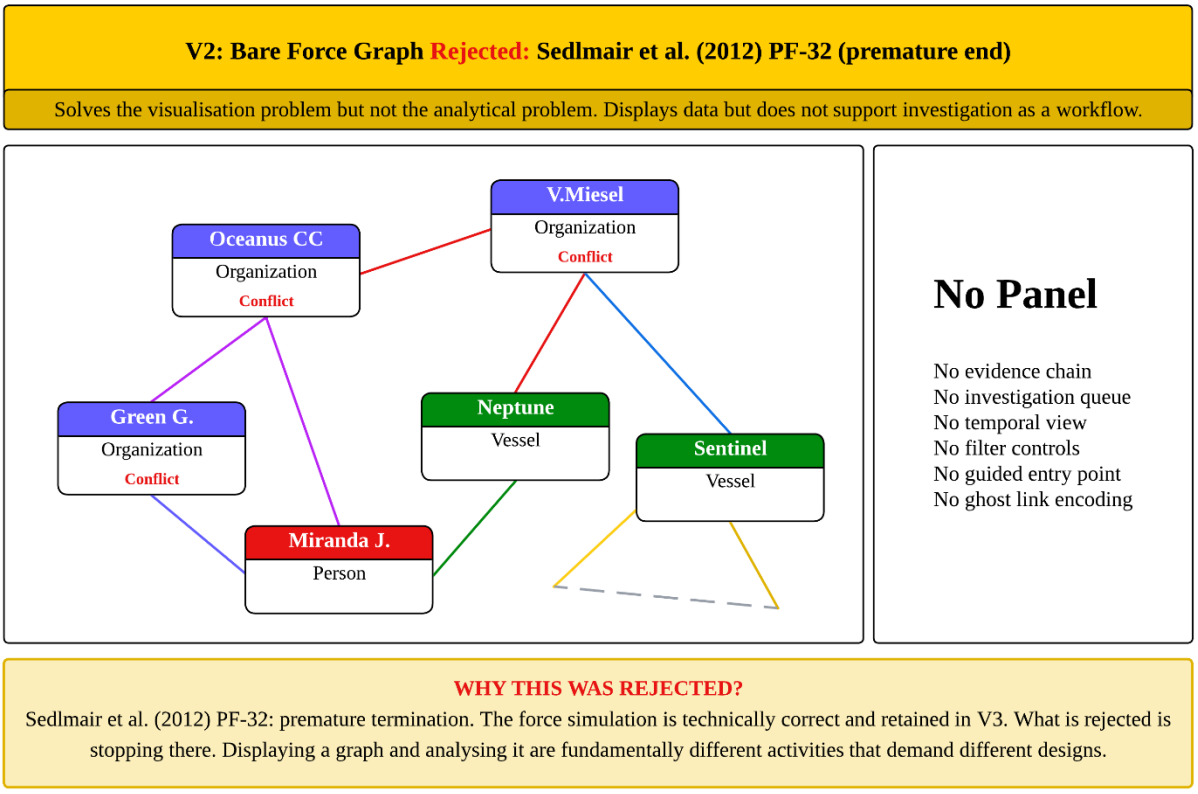

Figure 1c. V2: Bare force graph. Correctly spatialises the network but provides no investigation workflow. Rejected at Sedlmair et al. (2012) pitfall PF-32.

The third candidate was a standard D3 force-directed graph: the correct spatial representation for the network structure, but used in isolation without supporting workflow components. The rejection maps directly onto Sedlmair et al.'s pitfall PF-32, premature end, which describes designs that solve the visualisation problem but not the analytical problem. Displaying a knowledge graph and analysing it are fundamentally different activities that demand different designs. The force graph displays data but provides no mechanism to surface communication message content, no temporal view of the 14-day investigation window, no way to distinguish confirmed relationships from predicted missing ones, and no entry point that guides the analyst toward the most analytically significant entity. Shneiderman's details-on-demand is entirely absent. The force simulation is technically correct and is retained in V3. What was rejected was the decision to stop there.

V3: EvidenceBoard, the (Selected) Design

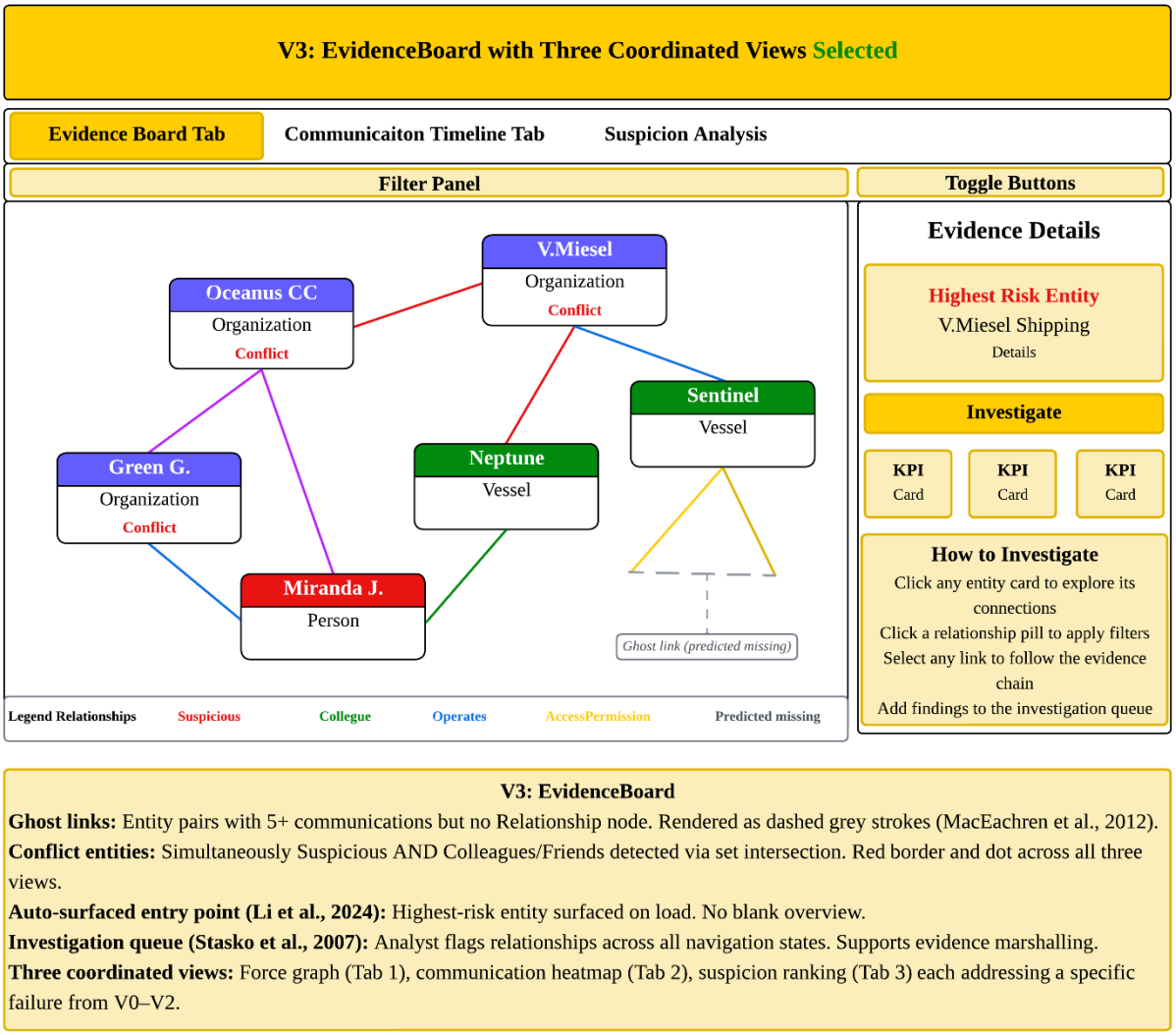

Figure 1d. V3: EvidenceBoard wireframe. Three coordinated views addressing each failure from V0 through V2. Ghost link shown as dashed stroke. Conflict entities marked with red border and dot.

The final design retains the force graph as the primary overview and addresses each specific failure of V2. The three-tab architecture implements Shneiderman's overview-and-detail hierarchy at a macro level: Tab 1 provides the entity-relationship network overview, Tab 2 provides a structured temporal overview, and Tab 3 provides a ranked analytical overview of the most suspicious entities. The investigation queue addresses Jigsaw's finding about external memory support. Ghost links address the knowledge graph incompleteness that V0, V1, and V2 all ignored. The auto-surfaced highest-risk entity on load responds directly to Li et al.'s finding that analysts prefer seeded entry points over blank overviews.

Applying Munzner's (2009) four-level nested model explicitly to EvidenceBoard: Level 1, the domain problem, is a coastal regulation analyst investigating suspicious permitting patterns in a knowledge graph too large and incomplete to read directly. Level 2, task abstraction, translates this into a primary Discover goal achieved through Browse and Explore, using Navigate, Filter,

and Select operations as defined by Brehmer and Munzner (2013). Level 3, visual encoding and interaction, maps entity type to colour hue, evidence count to stroke width, relationship confidence to stroke opacity, predicted missing connections to dashed texture, and conflict status to redundant red border and dot. Level 4, algorithm, is the D3 force simulation for spatial layout and set intersection for conflict detection. Each level's output directly constrains the level below it, as Munzner's model requires.

The Final Design

Visual Encoding: Every Channel Justified

The encoding decisions follow directly from Ware's (2004) effectiveness hierarchy and MacEachren et al.'s (2012) uncertainty guidelines.

Position encodes each entity's network role, determined by D3 force simulation with three competing forces: link distance scaled inversely with evidence count, charge repulsion preventing node overlap, and collision radius maintaining card legibility at any zoom level.

Colour hue encodes entity subtype and relationship type using a single consistent palette across all three tabs, a design discipline Ware identifies as critical for multi-view systems to enable cross-view recognition without requiring the analyst to re-read a legend.

Stroke width encodes evidence count on a compressed scale from 2 to 10 pixels, exploiting pre-attentive sensitivity to line weight over colour saturation.

Stroke opacity encodes relationship confidence, following MacEachren et al.'s empirical finding that opacity and colour value are the most intuitively understood uncertainty channels, rated 5.3 out of 7.0 in their experiment.

Dashed texture marks ghost links following cartographic convention for uncertain or provisional features, exploiting a widely understood visual convention rather than introducing a novel encoding.

Conflict entities receive a red border and filled red dot in their card header corner, providing a redundant encoding in both hue and shape as recommended by Ware for signals that must not be missed.

Tab 1: The EvidenceBoard

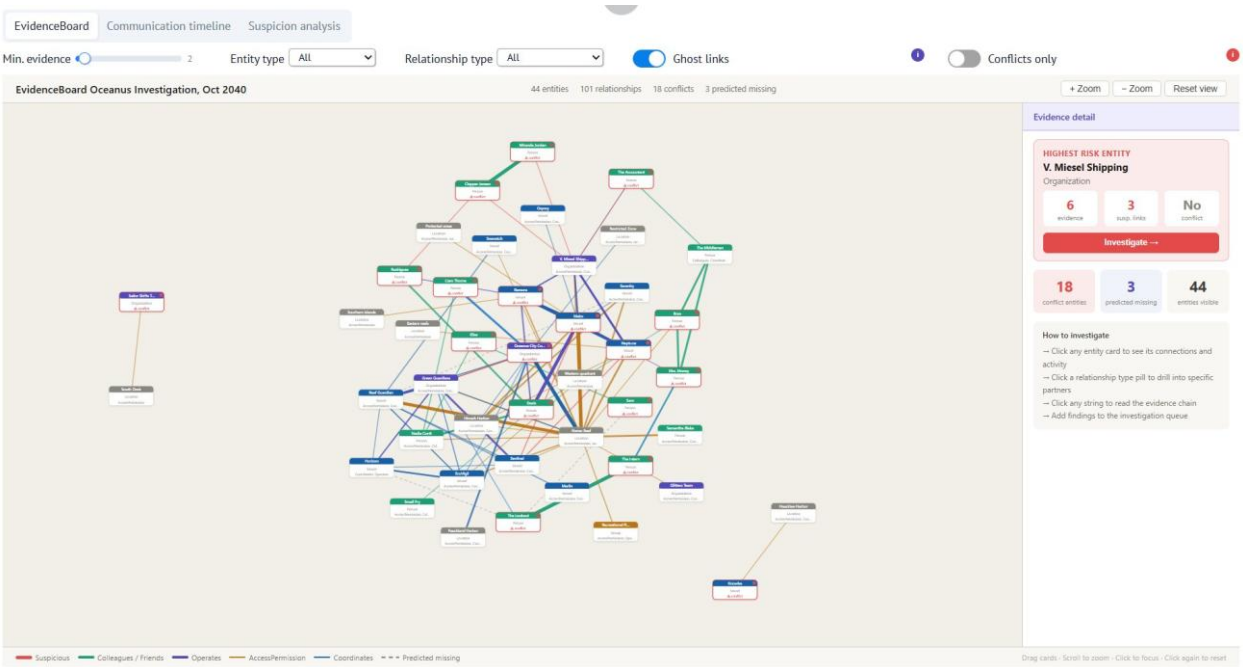

Figure 2. EvidenceBoard on first load. V. Miesel Shipping auto-surfaced as highest-risk entity. 44 entities, 18 conflicts, 3 ghost links visible.

The central tab implements Munzner's focus-and-context technique: clicking any entity card triggers focus mode, dimming non-connected entities to 10% opacity. Progressive disclosure operates across two layers. Layer 1 opens the entity detail panel showing a 14-day activity sparkline, sent and received message statistics, and relationship type pills. Layer 2 opens a mini star network for a specific relationship type, a ranked partner list with communication excerpts, and a back-navigation button. This implements Brehmer and Munzner's multi-level task chain: Discover, then Explore, then Identify.

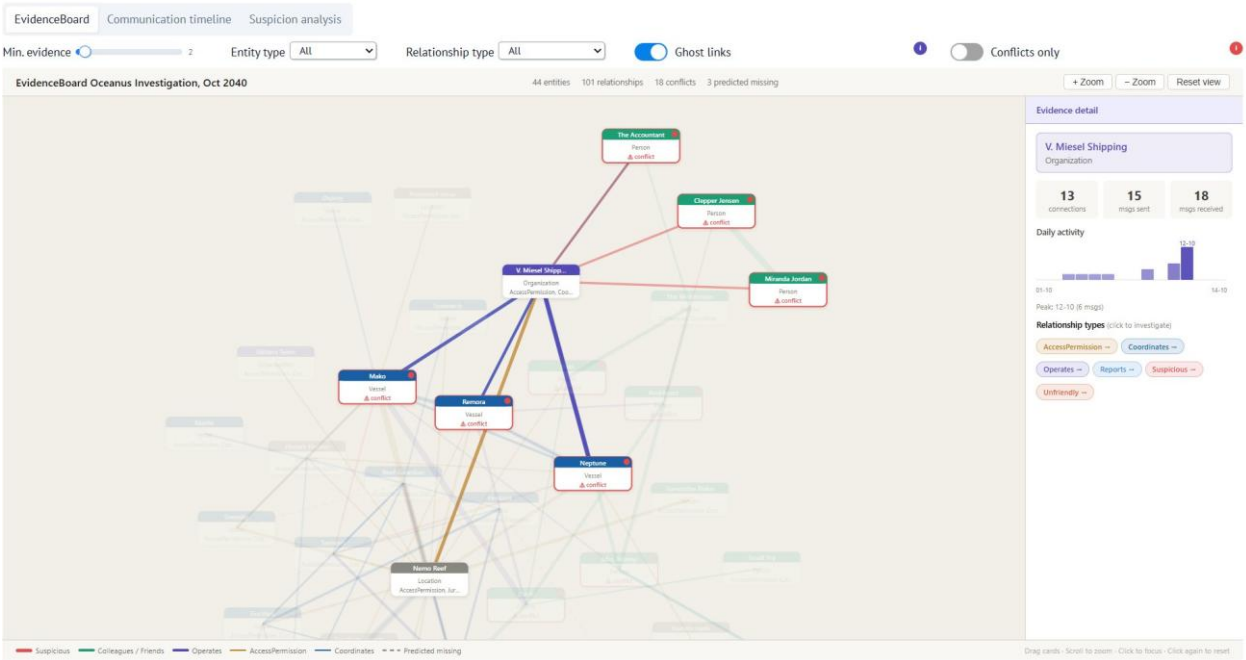

Figure 3. Focus mode on V. Miesel Shipping. Non-connected entities dimmed to 10% opacity. Evidence panel shows 13 connections and daily activity sparkline.

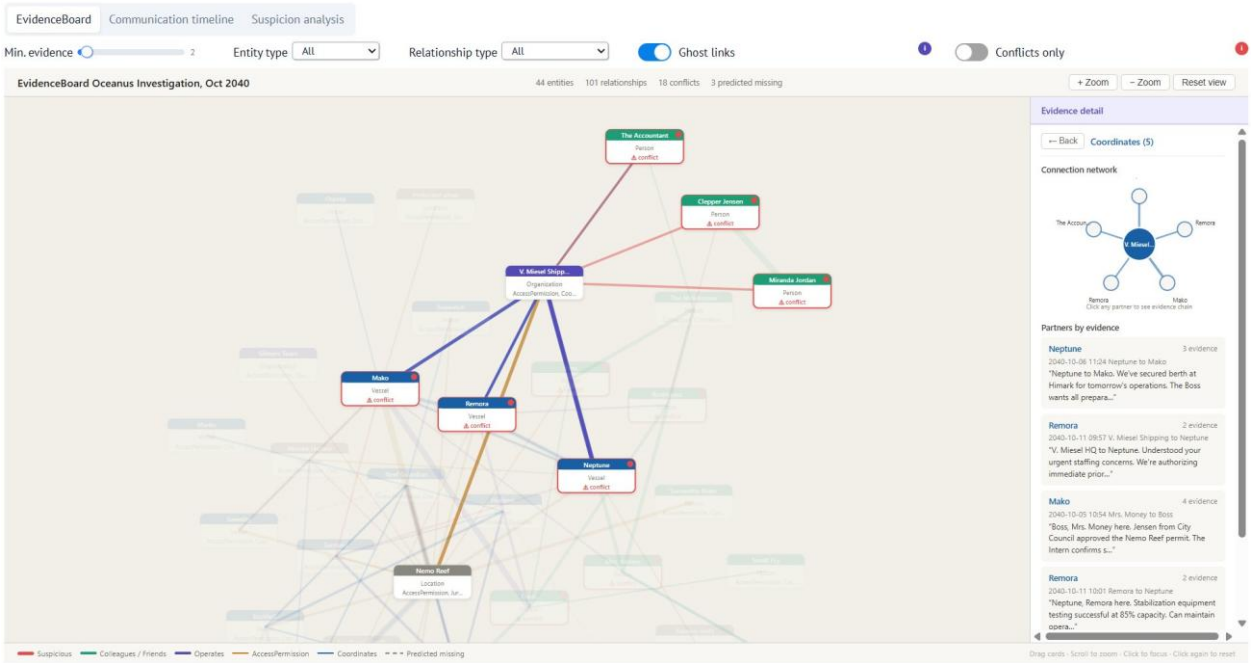

Figure 4. Layer 2 disclosure. Coordinates pill selected. Five partners ranked by evidence in a mini star network. Implements Brehmer and Munzner (2013): Discover, Explore, Identify.

The investigation queue allows the analyst to flag relationships by clicking Add to queue in the evidence chain panel as showed below. Flagged relationships persist across all navigation and filter changes and can be removed individually, implementing the evidence marshalling that Stasko et al. (2007) identified as essential for investigative workflows.

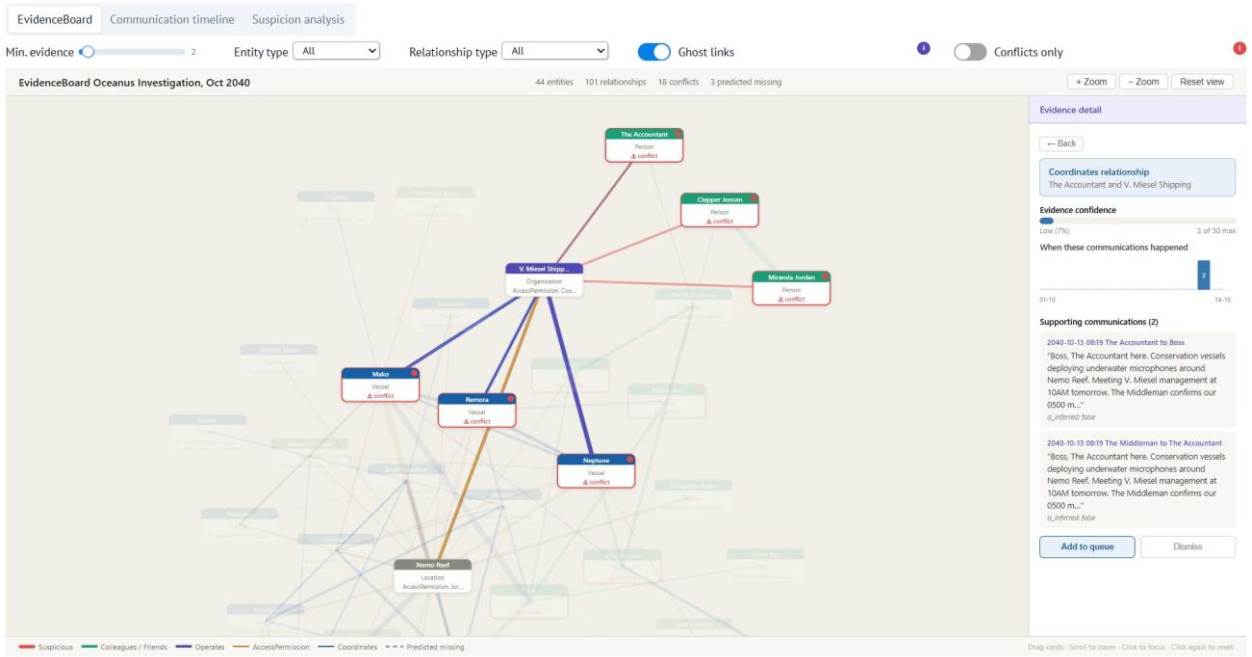

Figure 5. Evidence chain panel. Coordinates relationship between The Accountant and V. Miesel Shipping. Burst chart shows concentration on 13 October.

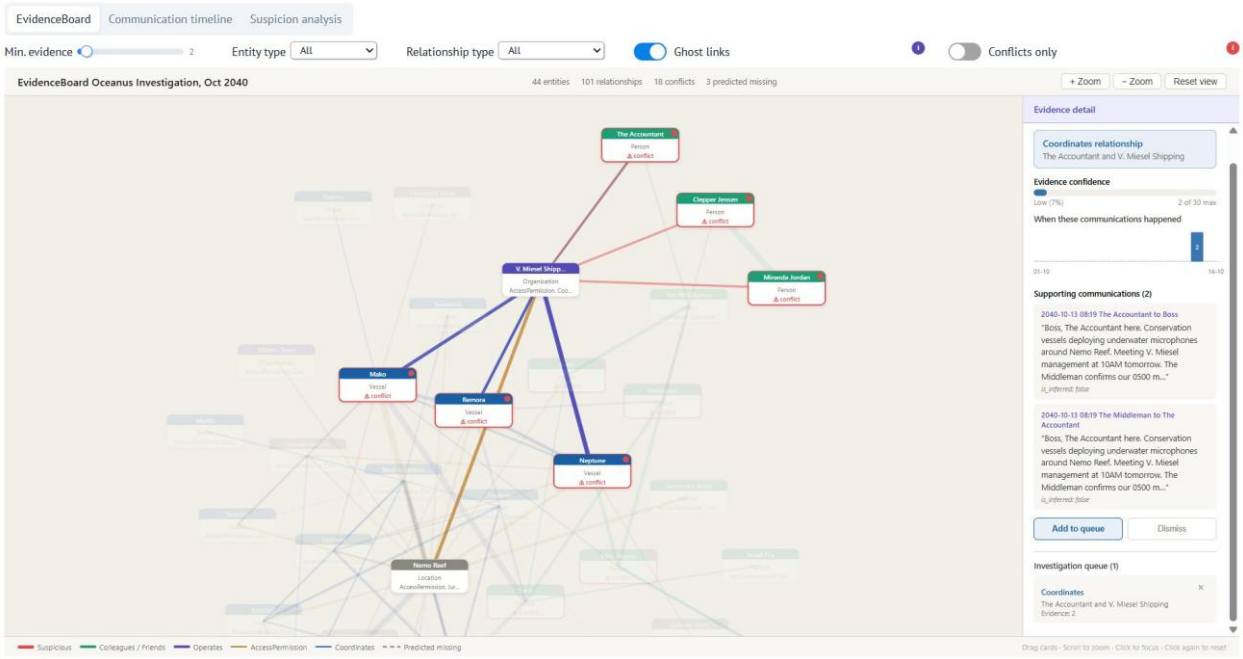

Figure 6. Investigation queue with one flagged relationship. Persists across all filter and navigation changes (Stasko et al., 2007).

Tab 2: Communication Timeline

Figure 7. Communication Timeline. Green Guardians on 06-10 selected. Morning activity concentration visible in hour histogram. Cross-highlight active.

The timeline view implements Shneiderman's (1996) overview-and-detail pattern for the temporal dimension. A heatmap matrix places the top 25 entities by message volume as rows and the 14 investigation days as columns. Cell colour uses a luminance-ordered sequential scale from light off-white to dark teal, following Ware's (2004) principle that ordered quantitative data should use ordered visual channels.

The two-colour cross-highlight, row in green and column in blue, allows the analyst to simultaneously compare an entity's activity across dates and compare all entities' activity on a specific day, answering two distinct analytical questions in a single hover gesture. Clicking a cell opens an hour-of-day histogram revealing whether communications concentrated in specific time windows. Bludau et al. (2023) argue that temporal provenance is a core dimension of epistemic uncertainty in knowledge graphs; the hour-of-day histogram directly implements this at sub-daily granularity.

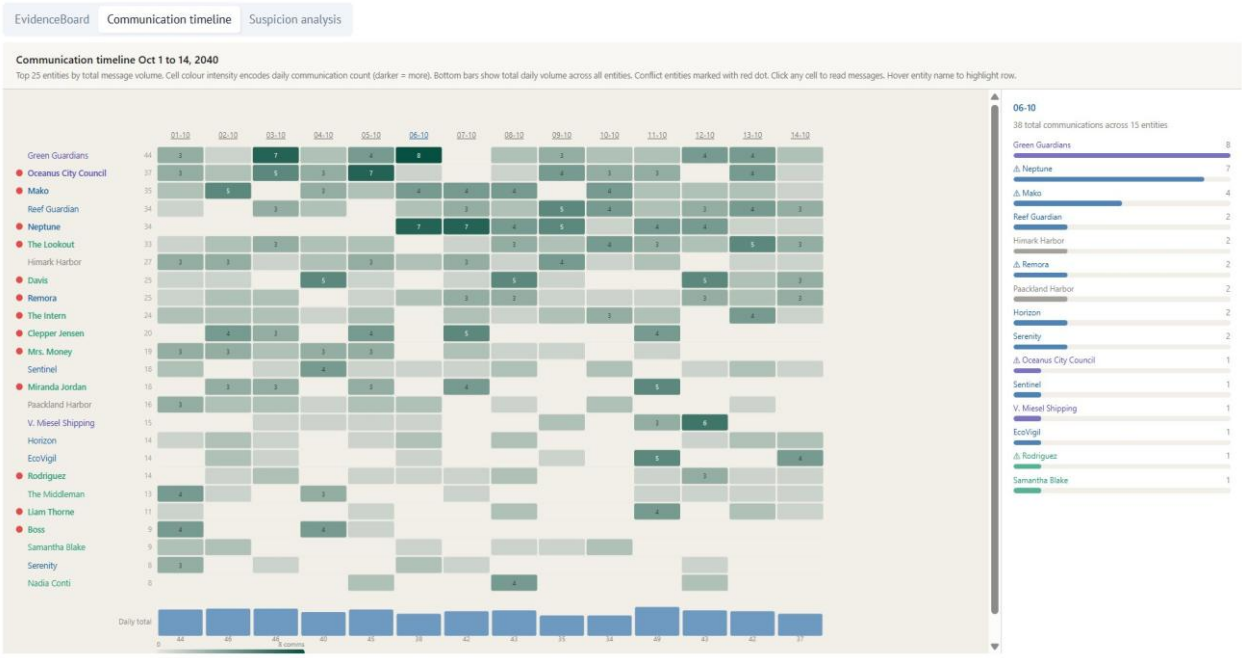

Figure 8. Column 06-10 selected. 38 total communications across 15 entities. Green Guardians (8), Neptune (7) and Mako (4) dominate.

Tab 3: Suspicion Analysis

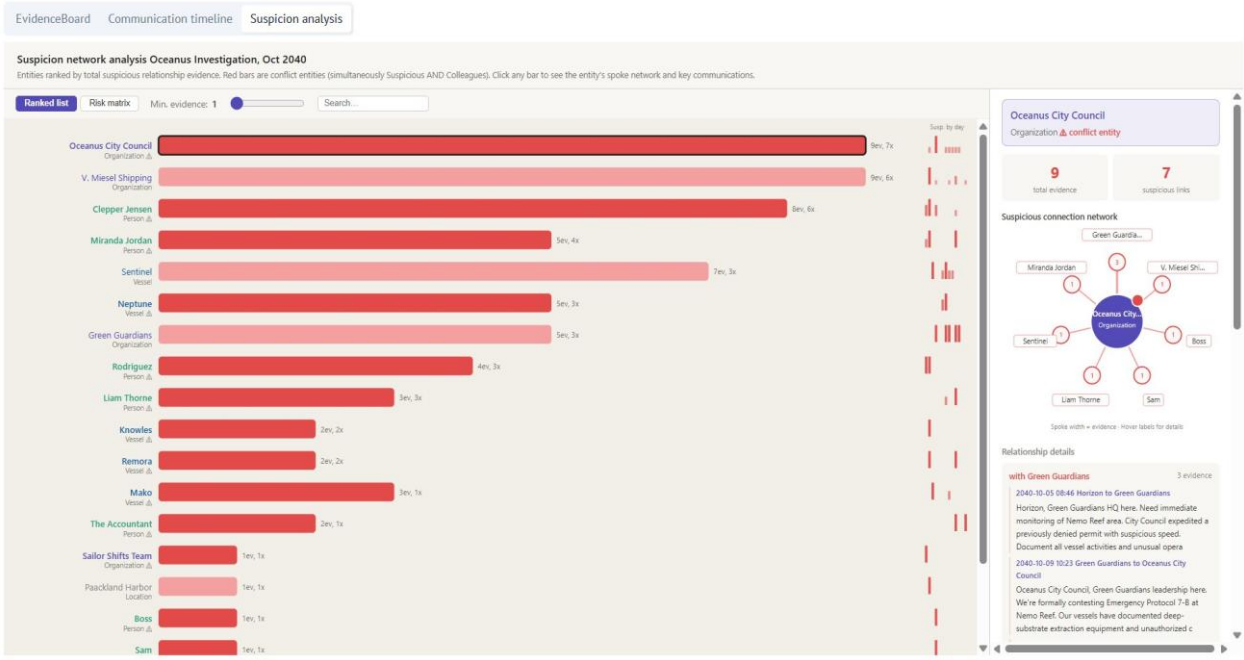

Figure 9. Suspicion Analysis. Oceanus City Council selected: 9 evidence, 7 suspicious links. Conflict badge visible. Spoke network shows all suspicious partners.

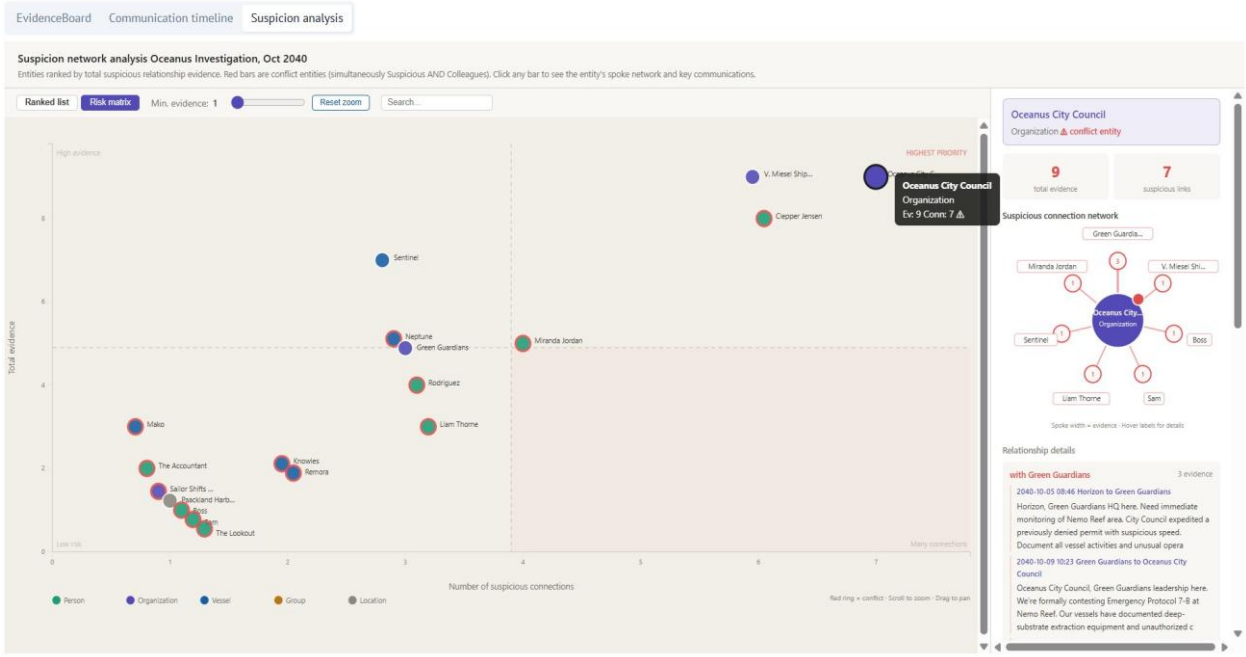

Figure 10. Risk matrix. Oceanus City Council in the HIGHEST PRIORITY quadrant with 9 evidence and 7 connections. Conflict ring in red.

The Suspicion Analysis tab provides two complementary views of the same analytical question: *which entities warrant the most investigative attention?* The ranked bar chart orders all entities with at least one Suspicious relationship by total evidence count. Conflict entities appear in solid red and non-conflict suspicious entities in lighter pink, a redundant encoding using both hue and value for reinforced discrimination. Next to each bar, a 14-day mini-sparkline shows the temporal distribution of suspicious evidence, implementing Brehmer and Munzner's (2013) juxtapose interaction. The risk matrix scatter plot places entities in a two-dimensional analytical space where horizontal position encodes number of suspicious connections and vertical position encodes total evidence. This decomposition reveals entities that a purely evidence-ranked list would obscure. The quadrant annotation HIGHEST PRIORITY in the top-right exploits natural reading direction to make the analytical conclusion immediately visible.

A Walkthrough: From Overview to Individual Message

Consider Nadia, a coastal regulation analyst with no graph theory background, investigating anomalous permitting activity around Nemo Reef in October 2040. She opens EvidenceBoard. Without clicking anything, the auto-surfaced panel identifies V. Miesel Shipping as the highest-risk entity with 6 evidence and 3 suspicious links. She clicks Investigate.

The board enters focus mode around V. Miesel Shipping. She sees six relationship type pills: AccessPermission, Coordinates, Operates, Reports, Suspicious, Unfriendly. She clicks Coordinates. Five partners appear in the mini star network: Neptune with 3 evidence, Remora

with 2, Mako with 4, and The Accountant with 2. She clicks Neptune and the evidence chain opens showing a temporal burst concentrated on 6 October. She flags this relationship to the investigation queue.

She switches to the Communication Timeline. Clicking column header 06-10 reveals 38 total communications across 15 entities that day. Green Guardians with 8, Neptune with 7 and Mako with 4 dominate. She notes Oceanus City Council shows a conflict badge. She switches to Suspicion Analysis, sets minimum evidence to 5, and toggles to the risk matrix. Oceanus City Council occupies the far top-right corner with 9 evidence and 7 connections. The complete workflow, from first load to queued findings, required fewer than twelve interactions.

Novel Contributions

Three design contributions distinguish EvidenceBoard from prior work.

Li et al. (2024) found that practitioners navigating knowledge graphs preferred to begin from a known, analytically grounded entity rather than a blank graph overview. Ghost link detection extends this principle to the edges themselves: the absence of a relationship node is no longer invisible, it is computationally derived, visually encoded, and immediately investigable.

Ghost link detection formalises the absence of a relationship node between two frequently communicating entities as a computationally derived prediction, rendering it as a first-class visual element with its own information panel and epistemic uncertainty encoding. No prior knowledge graph visualisation system reviewed here treats edge absence as a computationally encoded visual signal. The threshold of five communications is a heuristic; a production system would replace this with probabilistic link prediction providing ranked candidate missing relationships with associated confidence scores.

Conflict entity formalisation identifies entities simultaneously holding Suspicious and Colleagues or Friends relationships via set intersection, propagating this signal consistently through all three views. The three-view coordinated architecture extends Jigsaw's co-occurrence model into typed semantic relationships with quantified evidence chains and explicit temporal structure.

Generalisability

EvidenceBoard's three-view architecture is generalisable to any knowledge graph with typed relationships and embedded communication events. The force graph, heatmap, and suspicion ranking views make no assumptions specific to the MC3 dataset beyond the presence of entity nodes, relationship nodes, and timestamped communication nodes.

Several design elements are more tightly coupled to the MC3 dataset and would require re-parameterisation for a new domain. Ghost link detection assumes a specific edge vocabulary encoding sender and receiver provenance. Conflict entity detection is parameterised on the relationship labels Suspicious, Colleagues, and Friends, and a new domain would require identifying semantically equivalent opposing types. The heatmap timeline assumes a bounded investigation window of known duration. None of these are fundamental barriers to generalisation, but each requires deliberate mapping work before the design can operate on a different knowledge graph.

Scale is the most significant structural constraint on generalisability. Testing indicates the D3 force simulation degrades above approximately 150 entities. Li et al. (2024) found that practitioners typically navigate graphs with thousands of entities, meaning a production deployment would require server-side layout pre-computation and progressive entity loading rather than client-side force simulation.

Limitations

Data dependency. The current implementation is tightly bound to the MC3 label vocabulary and graph structure throughout. There is no configuration layer, no schema mapping interface, and no validation step that would alert a user if the loaded graph is missing required edge types or timestamp fields. If deployed with a structurally incompatible dataset, the ghost link detection and burst chart features would silently produce empty results rather than raising an informative error. A production system would require explicit schema validation, a configuration interface for remapping edge type labels, and graceful degradation when expected structural features are absent.

Environment constraints. The drag, hover, and precision click interactions assume a pointer device and a minimum screen width of approximately 1,280 pixels. The three-column layout does not adapt to mobile or tablet viewports. Touch-based devices would require a fundamentally different interaction model because hover is unavailable, drag targets require larger hit areas, and the evidence chain panel requires simultaneous screen space that a phone display cannot provide. The design is optimised for desktop environments with a large monitor.

Heuristic rather than probabilistic inference. Ghost link detection uses a threshold of five communications rather than a ranked probabilistic model. A richer implementation would score candidate missing relationships with confidence estimates.

No multi-analyst collaboration. The investigation queue is local to a single browser session. Supporting collaborative investigation across multiple analysts would require a shared backend.

Ethical Consideration

EvidenceBoard is designed around a fictional dataset, but the interaction patterns it encodes, surfacing individuals as highest-risk entities, flagging people as conflict entities, and predicting missing relationships between named persons, carry real ethical weight if applied to non-fictional contexts. Three specific risks warrant acknowledgement. First, the ghost link threshold of five communications is a heuristic, not a statistically validated model. Presenting a predicted missing relationship as a visual signal with the same visual weight as a confirmed one risks overstating the certainty of an inference. Second, the conflict entity label, applied to anyone simultaneously holding Suspicious and Colleagues or Friends relationships, is a computed signal derived from graph structure alone. It does not account for context, intent, or the reliability of the original relationship classification. A real deployment would require human review before any such label is acted upon. Third, the investigation queue allows an analyst to accumulate flagged relationships across a session. In a real investigative context this constitutes a record of suspicion that would require appropriate data governance, access controls, and retention policies. The current implementation has none of these safeguards, which is appropriate for a research prototype but would need to be addressed before any production deployment.

Implementation

EvidenceBoard is implemented as a single Marimo application (approximately 1,230 lines) structured in four reactive cells. Cell 1 imports Marimo. Cell 2 performs all data loading and Python computation: MC3 graph JSON traversal, entity and relationship extraction via type filtering, conflict entity detection via set intersection across relationship subtype sets per entity, ghost link identification by communication frequency thresholding at five communications per entity pair, sparkline statistics, and sender and receiver lookups for all 3,226 edges. Cell 3 creates the Marimo UI controls (minimum evidence slider, entity type dropdown, relationship type dropdown, ghost links toggle, and conflicts-only toggle). Cell 4 is the main reactive cell: it receives all controls and computed data, filters entities and relationships, serialises filtered data to JavaScript variables, and assembles three complete HTML iframe documents injected with D3.js v7 dynamically loaded from the public CDN.

D3.js is the rendering engine for all three tabs. Tab 1 uses forceSimulation with forceLink (link distance inversely scaled by evidence count), forceManyBody (charge repulsion at minus 300), forceCenter, and forceCollide (collision radius matching card dimensions), plus d3.zoom and d3.drag for pan, zoom, and drag interactions. Tab 2 uses scaleSequential with a custom luminance interpolator from off-white to dark teal. Tab 3 uses scaleLinear for both the ranked bar chart and the scatter plot, with zoom transform rescaling both axes on each zoom event so tick values remain readable at any zoom level.

Full source code: **github.com/SainiAmandeepSingh/EvidenceBoard.** Tools used: D3.js v7.8.5, Marimo v0.22.4, Python 3.

On first load, the interface requires approximately 30 to 60 seconds to initialise. This is expected behaviour as Marimo executes the full data processing pipeline in Cell 2 on startup, traversing all 3,226 edges and computing conflict entities, ghost links, and sparkline statistics before the interface renders. D3.js v7 additionally loads from the public CDN on first render of each tab. Subsequent filter and navigation interactions are immediate as all computation is cached in Marimo's reactive cell state.

Reflection

Working Without a Pre-Defined Task

Working without a specified analytical task sharpened the design process in an unexpected way. The shift from V2, a force graph that displayed data, to V3, a system that supported investigation, required explicitly acknowledging that displaying a knowledge graph and analysing it are fundamentally different activities that demand different designs. The most valuable methodological outcome was developing the habit of asking what specific task does this interaction serve before implementing any feature. This operationalises Brehmer and Munzner's (2013) why-how-what typology as a design filter. Features that survived the filter, including the investigation queue, the conflict badge, the burst chart, and the auto-surfaced insight panel, each support a specific and nameable analytical operation. Features that were contemplated but rejected, including cross-tab entity linking via shared state and animated temporal playback, failed the filter because their benefit was aesthetic rather than analytical.

The Most Difficult Aspect

The most difficult aspect was the combination of epistemic uncertainty and the decision to go beyond a conceptual design and build a working implementation.

On the design side, making uncertainty a first-class visual element through opacity encoding, dashed textures, and ghost links as analytical hypotheses required committing to a complex encoding scheme that risked confusing rather than informing. MacEachren et al.'s (2012) finding that opacity and dashed texture are among the most intuitively understood uncertainty channels, rated above colour saturation, provided the grounding that made this defensible.

On the implementation side, building a fully functional system in D3.js v7 within Marimo introduced constraints not present in a conceptual submission. Marimo prohibits accessing UI element values in the cell that defines them, requires iframes to be complete self-contained HTML documents, and executes JavaScript only inside iframe contexts. D3.js had to be loaded as a blocking script in the document head to prevent render race conditions. Each constraint required systematic debugging before the interface rendered correctly.

Acknowledging both challenges honestly rather than claiming either was straightforward reflects Sedlmair et al.'s (2012) caution against overstating the ease of any design study.

References

All DOIs link directly to the publisher record or IEEE Digital Library page.

Bludau, M.-J., Brüggemann, V., Burmester, A., and Dörk, M. (2023). Towards a relational perspective on knowledge graphs in digital humanities. Computer Graphics Forum, 42(3), e14831. https://doi.org/10.1111/cgf.14831

Brehmer, M., and Munzner, T. (2013). A multi-level typology of abstract visualization tasks. IEEE Transactions on Visualization and Computer Graphics, 19(12), 2376-2385. https://doi.org/10.1109/TVCG.2013.124

Li, H., Appleby, G., Brumar, C. D., Chang, R., and Suh, A. (2024). Knowledge graphs in practice: Characterizing their users, challenges, and visualization opportunities. IEEE Transactions on Visualization and Computer Graphics, 30(1), 584-594. https://doi.org/10.1109/TVCG.2023.3326904

MacEachren, A. M., Roth, R. E., O'Brien, J., Li, B., Swingley, D., and Gahegan, M. (2012). Visual semiotics and uncertainty visualization: An empirical study. IEEE Transactions on Visualization and Computer Graphics, 18(12), 2496-2505. https://doi.org/10.1109/TVCG.2012.252

Munzner, T. (2009). A nested model for visualization design and validation. IEEE Transactions on Visualization and Computer Graphics, 15(6), 921-928. https://doi.org/10.1109/TVCG.2009.111

Sedlmair, M., Meyer, M., and Munzner, T. (2012). Design study methodology: Reflections from the trenches and the stacks. IEEE Transactions on Visualization and Computer Graphics, 18(12), 2431-2440. https://doi.org/10.1109/TVCG.2012.213

Shneiderman, B. (1996). The eyes have it: A task by data type taxonomy for information visualizations. Proceedings of the IEEE Symposium on Visual Languages, 336-343. https://doi.org/10.1109/VL.1996.545307

Stasko, J., Görg, C., and Liu, Z. (2007). Jigsaw: Supporting investigative analysis through interactive visualization. Proceedings of IEEE VAST 2007, 131-138. https://doi.org/10.1109/VAST.2007.4389006

Ware, C. (2004). Information Visualization: Perception for Design (2nd ed.). Morgan Kaufmann. ISBN: 978-1-55860-819-1. Note: books do not carry DOIs; the publisher record is accessible at https://www.sciencedirect.com/book/9781558608191/information-visualization

EvidenceBoard was built with Marimo v0.22.4, D3.js v7.8.5, and Python 3.

AI Use Statement

In accordance with Utrecht University AI Policy Level 4, this project used AI assistance in the following ways.

Claude (Anthropic) was used for implementation debugging throughout the development of the Marimo application. Specific uses included diagnosing why JavaScript failed to execute inside mo.Html() contexts, resolving Marimo cell dependency errors caused by accessing UI element values in the same cell that defined them, and identifying the correct iframe architecture for injecting D3.js v7 into a Marimo environment. The prompts consisted of specific error messages and code snippets. The outputs were diagnostic explanations and corrected code fragments, all of which were reviewed, tested, and integrated manually.

ChatGPT (OpenAI) was used for light language refinement in selected sections of this report. All content, including design decisions, literature selections, analytical arguments, and conclusions, was written by the author. ChatGPT was used solely to refine phrasing and improve academic tone in places where the author had already written the substantive content. The author is fully responsible for the accuracy and quality of all submitted content.